TAPER MILL

CP AND CT TAPER MILLS

MILLING

GENERAL DESCRIPTION

TAPER MILLS REMOVE RESTRICTIONS AND MILL THROUGH PINCHED OR COLLAPSED CASING. THEY CARRY A TAPERED OR SHORT BLUNT NOSE THAT SERVES AS A GUIDE. TWO VERSIONS: THE CP, WITH A SHORT BLUNT NOSE AND A SHORT TAPER, PRODUCES MUCH LESS TORQUE AND CAN TAKE MORE WEIGHT, SUITING THE TOUGHEST JOBS WHERE A LONG NOSE MIGHT BREAK; THE CT, WITH SPIRAL BLADES AND A POINTED NOSE, IS DESIGNED TO MILL THROUGH RESTRICTIONS.

TECHNICAL CHARACTERISTICS

Dressed O.D.	Top pin conn. (API reg.)	Length CT	Length CP	Fishing neck length	Fishing neck O.D.	Wt CT (lb)	Wt CP (lb)
3-1/4 - 3-7/8	2-3/8	34	30	10	3	80	60
4 - 4-3/8	2-3/8	34	30	10	3-1/8	90	70
4-1/2 - 5-3/8	2-7/8	38	31	10	3-3/4	106	75
5-1/2 - 5-5/8	3-1/2	42	32	13	4-1/4	155	115
5-3/4 - 6-3/8	3-1/2	44	32	13	4-3/4	160	120
6-1/2 - 7-3/8	3-1/2	46	34	13	4-3/4	170	130
7-1/2 - 7-7/8	4-1/2	54	36	13	5-3/4	250	185
8-3/8 - 9-1/2	4-1/2	54	36	14	5-3/4	280	220
9-3/8 - 9-7/8	4-1/2 or 6-5/8	54	36	14	5-3/4 or 7-3/4	345	280
10-3/8 - 11	6-5/8	57	38	14	7-3/4	415	355
11-1/2 - 12-1/4	6-5/8	60	40	14	7-3/4	455	390
14-3/4 - 15	6-5/8	70	60	18	7-3/4	525	460
17 - 17-1/2	6-5/8	70	60	18	7-3/4	595	530
20 - 26	6-5/8 or 7-5/8	76	66	18	7-3/4 or 9-1/2	1,250	1,125

Dimensions in inches (overall lengths in inches), weights in lb (approximate). Ranges as dressed diameters of each size group.

DESIGN FEATURES

- **CP:** blunt nose, short taper; low torque, tolerates more weight.
- **CT:** spiral blades and pointed nose for restrictions.
- **Typical jobs:** reaming collapsed casing and liners, cleaning permanent whipstock windows, milling jagged or split shoes, enlarging restrictions through retainers and adapters.
- **Stabilising:** keep 1/2 to 1 in. under drift diameter.

OPERATING PARAMETERS

Weight range (general)	2,000 - 4,000 lb
Start rotation above collapse	75 rpm (min. 75 rpm)
Weight increase in collapse	1,000 - 2,000 lb, slowly
Collapsed casing	about 50 rpm; 2,000 - 3,000 lb
Whipstock window clean-up	40 rpm; under 1,000 lb
Retainers / adapters	about 70 rpm; 2,000 - 6,000 lb
After clearing restriction	80 - 100 rpm

OPERATING NOTES

Run a mill about 1/4 in. larger than the minimum I.D. of the collapse and enlarge in stages; do not use a taper mill if the collapse has passed centre. A string taper mill can be used if there is a danger of sidetracking. Governing factor is torque: reduce speed and weight if it rises. For a window, run the same diameter as the largest window mill, rotate slowly its full length and repeat until torque is minimal; do not try to make hole.

HOW TO ORDER

Please specify: 1. type (CP or CT) 2. dressed diameter 3. top pin connection 4. fishing neck diameter, if different from standard.